

Exposé III. Functoriality of Categories of Sheaves

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Translator's status note. This is a working English translation of the complete SGA 4, Exposé III. The source is the Orgogozo–Laszlo TeX transcription, file 03/03.tex, lines 1–1551. Notation is kept close to the French original: $\widehat{C}$ denotes presheaves on C , $\widetilde{C}$ denotes sheaves on C , $\underline{a}$ is sheafification, and $\epsilon_C : C \rightarrow \widetilde{C}$ is the canonical functor. Terminology is normalized to current English usage: *sieve*, *covering sieve*, *bicovering morphism*, *continuous functor*, *cocontinuous functor*, *induced topology*, and *comparison lemma*.

In Exposé I, Section 5, the behavior of categories of presheaves with respect to functors between the categories of arguments was studied. In this exposé that study is extended to the case of sites and categories of sheaves (Sections 1 and 2). After introducing the induced topology (Section 3), we turn in Section 4 to the comparison lemma, which will play an important role in Exposé IV. In Section 5, for the patient reader, certain commutative diagrams connected with localized categories of the type C/X are studied. In Theorem 5.1 of Exposé I one imposes smallness hypotheses on the categories under consideration. These hypotheses are not, in general, satisfied by the sites occurring in practice ($\mathcal{U}$ -sites). This forces some contortions in 4.2, 4.3, and 4.4.

1 Continuous functors

Definition 1.1. Let C and C' be two $\mathcal{U}$ -sites, and let $u : C \rightarrow C'$ be a functor between the underlying categories. We say that u is *continuous* if, for every sheaf of sets F on C' , the presheaf

$$X \longmapsto F \circ u(X)$$

on C is a sheaf.

This notion of continuity depends *a priori* on the universe $\mathcal{U}$ for which the two sites are $\mathcal{U}$ -sites. Proposition 1.5 shows that in fact it does not.

1.1.1

Let

$$i : \widetilde{C} \longrightarrow \widehat{C}, \quad i' : \widetilde{C'} \longrightarrow \widehat{C'}$$

denote the canonical inclusion functors of sheaves into presheaves. By Definition 1.1, the functor u is continuous if and only if there exists a functor

$$u_s : \widetilde{C'} \longrightarrow \widetilde{C}$$

such that

$$iu_s = u^*i', \quad u^*F = F \circ u$$

(Exposé I, 5.0).

Proposition 1.2. Let C be a small site, let C' be a $\mathcal{U}$ -site, and let $u : C \rightarrow C'$ be a functor between the underlying categories. The following properties are equivalent:

- i) The functor u is continuous.
- ii) For every object X of C and every covering sieve $R \hookrightarrow X$, the morphism

$$u_! R \hookrightarrow u(X)$$

is bicobering in $\widehat{C'}$ (Exposé II, 5.2 and Exposé I, 5.1).¹

- iii) For every bicobering family $H_i \rightarrow K$, $i \in I$, in $\widehat{C}$, the family

$$u_! H_i \longrightarrow u_! K, \quad i \in I,$$

is bicobering in $\widehat{C'}$.

- iv) There exists a functor

$$u^s : \widetilde{C} \longrightarrow \widetilde{C'}$$

commuting with inductive limits and “extending u ”, i.e. such that the diagram of canonical functors

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ \downarrow \epsilon_C & & \downarrow \epsilon_{C'} \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C'} \end{array}$$

is commutative up to isomorphism.

Moreover, when C is no longer assumed to be a small site, but is a $\mathcal{U}$ -site, one still has i) $\Leftrightarrow$ iv), and the functor u^s of iv) is necessarily left adjoint to

$$u_s : \widetilde{C'} \longrightarrow \widetilde{C};$$

in particular, it is determined up to unique isomorphism.

Proof. The proof of i) $\Leftrightarrow$ iv) when C is a $\mathcal{U}$ -site will be given in 4.2. Suppose first that C is small. It is clear that iii) $\Rightarrow$ ii).

i) $\Rightarrow$ iii). Let $H_i \rightarrow K$, $i \in I$, be a bicobering family in $\widehat{C}$. For every sheaf F on C' , the presheaf $u^* F$ is a sheaf on C . Hence, by Exposé II, 5.3, the canonical map

$$\mathrm{Hom}(K, u^* F) \longrightarrow \prod_i \mathrm{Hom}(H_i, u^* F)$$

is bijective. By adjunction (Exposé I, 5.1), it follows that

$$\mathrm{Hom}(u_! K, F) \longrightarrow \prod_i \mathrm{Hom}(u_! H_i, F)$$

is bijective. Thus, again by Exposé II, 5.3, the family $u_! H_i \rightarrow u_! K$, $i \in I$, is bicobering.

ii) $\Rightarrow$ i). Let X be an object of C , let $R \hookrightarrow X$ be a covering sieve, and let F be a sheaf on C' . By Exposé II, 5.3, the map

$$\mathrm{Hom}(u_! X, F) \longrightarrow \mathrm{Hom}(u_! R, F)$$

¹“Covering” in place of “bicobering” is not necessarily sufficient; see Example 1.9.3 below.

is bijective. By adjunction, it follows that

$$\mathrm{Hom}(X, u^*F) \longrightarrow \mathrm{Hom}(R, u^*F)$$

is bijective. Hence u^*F is a sheaf.

i) $\Rightarrow$ iv). Use the usual notation $\underline{a}$ and i (respectively $\underline{a}'$ and i') for the associated-sheaf functor and the inclusion into presheaves. For a sheaf G on C , put

$$u^s G = \underline{a}' u_! i G.$$

For every sheaf F on C' , one has natural isomorphisms

$$\begin{aligned} \mathrm{Hom}(G, u_s F) &\simeq \mathrm{Hom}(iG, i u_s F) \simeq \mathrm{Hom}(iG, u^* i' F) \\ &\simeq \mathrm{Hom}(u_! iG, i' F) \simeq \mathrm{Hom}(\underline{a}' u_! iG, F). \end{aligned}$$

The first isomorphism comes from the full faithfulness of i ; the second from the definition of u_s ; the third and fourth from adjunction. Thus u^s is left adjoint to u_s . In particular, u^s commutes with inductive limits.

For every presheaf K on C and every sheaf F on C' , the same adjunctions give natural isomorphisms

$$\mathrm{Hom}(u^s \underline{a} K, F) \simeq \mathrm{Hom}(\underline{a} K, u_s F) \simeq \mathrm{Hom}(K, i u_s F) \simeq \mathrm{Hom}(K, u^* i' F) \simeq \mathrm{Hom}(u_! K, i' F) \simeq \mathrm{Hom}(\underline{a}' u_! K, F).$$

Hence there is a functorial isomorphism

$$u^s \underline{a} K \simeq \underline{a}' u_! K.$$

Taking K representable, and using Exposé I, 5.4(3), one obtains a diagram commutative up to isomorphism:

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ \downarrow \epsilon_C & & \downarrow \epsilon_{C'} \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}'. \end{array}$$

iv) $\Rightarrow$ ii). Let $h : C \rightarrow \widehat{C}$ and $h' : C' \rightarrow \widehat{C}'$ be the Yoneda functors. Consider the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ \downarrow h & & \downarrow h' \\ \widehat{C} & \xrightarrow{u_!} & \widehat{C}' \\ \downarrow \underline{a} & & \downarrow \underline{a}' \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}'. \end{array}$$

The upper square is commutative (Remark 1.4(3)), and $\underline{a}h = \epsilon_C$, $\underline{a}'h' = \epsilon_{C'}$. For every object K of $\widehat{C}$, one has the canonical isomorphism

$$\varinjlim_{(h(X) \rightarrow K) \in \mathrm{Ob}(C/K)} h(X) \xrightarrow{\sim} K$$

(Exposé I, 3.4). Since $u_!$, u^s , $\underline{a}$, and $\underline{a}'$ commute with inductive limits, it follows that

$$\varinjlim_{(h(X) \rightarrow K)} u^s \underline{a} h(X) \xrightarrow{\sim} u^s \underline{a} K, \quad \varinjlim_{(h(X) \rightarrow K)} \underline{a}' u_! h(X) \xrightarrow{\sim} \underline{a}' u_! K.$$

Since $u_!h = h'u$, and by hypothesis $\underline{a}'h'u \simeq u^s\underline{a}h$, one obtains a functorial isomorphism

$$u^s\underline{a}K \simeq \underline{a}'u_!K.$$

The diagram

$$\begin{array}{ccc} \widehat{C} & \xrightarrow{u_!} & \widehat{C}' \\ \downarrow \underline{a} & & \downarrow \underline{a}' \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}' \end{array}$$

is therefore commutative up to isomorphism. Since $\underline{a}i$ is isomorphic to the identity functor, one has $u^s \simeq \underline{a}'u_!i$, giving uniqueness of u^s . If $v : H \rightarrow K$ is a bicovering morphism in $\widehat{C}$, then $\underline{a}(v)$ is an isomorphism (Exposé II, 5.3), and hence $\underline{a}'u_!(v)$, isomorphic to $u^s\underline{a}(v)$, is an isomorphism. Thus $u_!(v)$ is transformed by $\underline{a}'$ into an isomorphism, and consequently $u_!(v)$ is bicovering. This proves ii). The additional assertion in the small case was proved along the way. $\square$

1.2.1

The functor $u^s : \widetilde{C} \rightarrow \widetilde{C}'$ of Proposition 1.2(iv) can often be interpreted as an “inverse-image” functor for a “morphism of topoi” $\widetilde{C}' \rightarrow \widetilde{C}$; cf. Exposé IV, 4.9. Its properties are summarized in the following proposition.

Proposition 1.3. Let $u : C \rightarrow C'$ be a continuous functor from a small site C to a $\mathcal{U}$ -site C' .

- 1) The functor u^s is left adjoint to u_s .
- 2) There is a canonical isomorphism $u^s \simeq \underline{a}'u_!i$.
- 3) There is a canonical isomorphism $u^s\underline{a} \simeq \underline{a}'u_!$.
- 4) The functor u^s commutes with inductive limits.
- 5) When $u_!$ is left exact (cf. Exposé I, 5.4(4)), u^s is left exact as well. More generally, u^s commutes with the same finite projective-limit types with which $u_!$ commutes.

Proof. Assertions (1)–(4) were proved during the proof of Proposition 1.2. Assertion (5) follows from the isomorphism in (2), since i commutes with projective limits and $\underline{a}'$ commutes with finite projective limits (Exposé II, 4.1). $\square$

Remark 1.4. Combining Exposé I, 5.4(3), with Proposition 1.3(3), one obtains a diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ \downarrow h & & \downarrow h' \\ \widehat{C} & \xrightarrow{u_!} & \widehat{C}' \\ \downarrow \underline{a} & & \downarrow \underline{a}' \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}' \end{array}$$

whose upper square is commutative and whose lower square is commutative up to isomorphism. But the functors $\underline{a}$, $\underline{a}'$, $u_!$, and u^s are defined only up to isomorphism. The reader may check that the functors $\underline{a}$, $\underline{a}'$, and u^s can be chosen so that:

- 1) the composite functors $\underline{a}h$ and $\underline{a}'h'$ are injective on sets of objects;
- 2) the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ \downarrow \underline{a}h & & \downarrow \underline{a}'h' \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}' \end{array}$$

is commutative.

More precisely, one may first choose $\underline{a}$ and $\underline{a}'$ so as to satisfy (1). Once these have been chosen, one may choose u^s so as to satisfy (2).

Proposition 1.5. Let $\mathcal{U} \subset \mathcal{V}$ be two universes, let C and C' be two $\mathcal{U}$ -sites, and let $u : C \rightarrow C'$ be a functor between the underlying categories. Then u is continuous relative to $\mathcal{U}$ if and only if it is continuous relative to $\mathcal{V}$. Moreover, when u is continuous, let $u_{\mathcal{U}}^s$ (respectively $u_{\mathcal{V}}^s$) denote the functor between categories of $\mathcal{U}$ -sheaves (respectively $\mathcal{V}$ -sheaves) introduced in Proposition 1.2(iv). Then the diagram

$$\begin{array}{ccc} \widetilde{C}_{\mathcal{U}} & \xrightarrow{u_{\mathcal{U}}^s} & \widetilde{C}'_{\mathcal{U}} \\ \downarrow & & \downarrow \\ \widetilde{C}_{\mathcal{V}} & \xrightarrow{u_{\mathcal{V}}^s} & \widetilde{C}'_{\mathcal{V}} \end{array} \quad (1.5.1)$$

where the vertical functors are the canonical inclusions, is commutative up to canonical isomorphism.

Proof. We give the proof only when C is $\mathcal{U}$ -small. The general case will be proved in 4.3. It is clear that if u^* sends every $\mathcal{V}$ -sheaf on C' to a $\mathcal{V}$ -sheaf on C , then it sends every $\mathcal{U}$ -sheaf on C' to a $\mathcal{U}$ -sheaf on C . Conversely, suppose u^* sends every $\mathcal{U}$ -sheaf on C' to a $\mathcal{U}$ -sheaf on C . Write

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}, \quad \widehat{C}'_{\mathcal{U}} \hookrightarrow \widehat{C}'_{\mathcal{V}}$$

for the categories of $\mathcal{U}$ -presheaves and $\mathcal{V}$ -presheaves, and write $u_{\mathcal{U}!}$ and $u_{\mathcal{V}!}$ for the corresponding left Kan extension functors. By the explicit construction of $u_!$ in Exposé I, 5.1, one has a commutative diagram

$$\begin{array}{ccc} \widehat{C}_{\mathcal{U}} & \xrightarrow{u_{\mathcal{U}!}} & \widehat{C}'_{\mathcal{U}} \\ \downarrow & & \downarrow \\ \widehat{C}_{\mathcal{V}} & \xrightarrow{u_{\mathcal{V}!}} & \widehat{C}'_{\mathcal{V}} \end{array} \quad (*)$$

The inclusion functors from $\mathcal{U}$ -presheaves to $\mathcal{V}$ -presheaves are fully faithful and commute with projective limits. Hence, by Exposé II, 5.1(i), a biconverging morphism of $\mathcal{U}$ -presheaves is a biconverging morphism of $\mathcal{V}$ -presheaves. Now let $R \hookrightarrow X$ be a covering sieve of an object X of C . By Proposition 1.2(ii), $u_{\mathcal{U}!}R \rightarrow u_{\mathcal{U}!}X$ is biconverging. Using (*) and the preceding observation, we deduce that $u_{\mathcal{V}!}R \rightarrow u_{\mathcal{V}!}X$ is biconverging. Hence, by Proposition 1.2, u^* sends $\mathcal{V}$ -sheaves on C' to $\mathcal{V}$ -sheaves on C . The commutativity of (1.5.1) follows from the commutativity of (*), Exposé II, 3.6, and Proposition 1.3(2). $\square$

Proposition 1.6. Let C and C' be two $\mathcal{U}$ -sites, and let $u : C \rightarrow C'$ be a functor between the underlying categories. Consider the following conditions:

- i) u is continuous (cf. Definition 1.1 and Proposition 1.2).

ii) For every covering family $(X_i \rightarrow X)_{i \in I}$ in C , the family $(uX_i \rightarrow uX)_{i \in I}$ in C' is covering.

Then i) $\Rightarrow$ ii).

Suppose that the topology of C can be defined by a pretopology T (Exposé II, 1.3) and that u commutes with fiber products.² Then i) and ii) are equivalent, and are also equivalent to:

iii) The functor u sends covering families for T to covering families in C' .

In particular, suppose that in C fiber products are representable and that u commutes with fiber products. Then conditions i) and ii) are equivalent.

Proof. First prove i) $\Rightarrow$ ii). Let $(X_i \rightarrow X)_{i \in I}$ be a covering family in C . For every sheaf F on C' , u^*F is a sheaf. Hence

$$\mathrm{Hom}(X, u^*F) \longrightarrow \prod_i \mathrm{Hom}(X_i, u^*F)$$

is injective (Exposé II, 5.1). Therefore

$$\mathrm{Hom}(u_!X, F) \longrightarrow \prod_i \mathrm{Hom}(u_!X_i, F)$$

is injective. Thus the family $u_!X_i \rightarrow u_!X$, $i \in I$, is covering in C' .

Every covering family for the pretopology T is a covering family in C , so ii) $\Rightarrow$ iii). We prove iii) $\Rightarrow$ i). Let X be an object of C , and let $(X_i \rightarrow X)_{i \in I}$ be a covering family in $\mathrm{Cov}(X)$ (Exposé II, 1.3). The morphisms $X_i \rightarrow X$ are base-changeable and u commutes with the relevant fiber products. Hence the fiber products $u(X_i) \times_{u(X)} u(X_j)$ are representable and canonically isomorphic to $u(X_i \times_X X_j)$. Let F be a sheaf on C' . Then the diagram of sets

$$F(u(X)) \longrightarrow \prod_i F(u(X_i)) \rightrightarrows \prod_{i,j} F(u(X_i \times_X X_j))$$

is exact (Exposé II, 2.1; Exposé I, 3.5 and 2.12). Therefore

$$u^*F(X) \longrightarrow \prod_i u^*F(X_i) \rightrightarrows \prod_{i,j} u^*F(X_i \times_X X_j)$$

is exact. Hence u^*F is a sheaf (Exposé II, 2.4), proving i). The final assertion follows from the preceding argument and from the fact that, when fiber products are representable in C , all covering families of C , indexed by sets in a suitable universe, define a pretopology whose associated topology is the topology of C . $\square$

Proposition 1.7. Let C be a small site, let C' be a $\mathcal{U}$ -site, and let $u : C \rightarrow C'$ be continuous. Let γ be an algebraic structure defined by finite projective limits, such that in the category of γ -objects in $\mathcal{U}\text{-Set}$ the $\mathcal{U}$ -inductive limits are representable.³ We use the notation of Exposé II, 6.4. The functor u_s commutes with projective limits and therefore defines a functor

$$u_s^\gamma : \tilde{C}_\gamma \longrightarrow \tilde{C}'_\gamma.$$

The functor u_s^γ has a left adjoint u_γ^s , with the following properties:

²In fact it is enough that u commute with the fiber products occurring in base change for morphisms coming from covering families for T .

³In fact one can show that this condition is always satisfied.

1) There is a canonical isomorphism

$$u_\gamma^s \xrightarrow{\sim} \underline{a}'_\gamma u_{\gamma!} i_\gamma,$$

and u_γ^s commutes with the finite projective limits with which $u_{\gamma!}$ commutes.

2) There is a canonical isomorphism

$$u_\gamma^s \underline{a}_\gamma \xrightarrow{\sim} \underline{a}'_\gamma u'_{\gamma!}.$$

3) The functor u_γ^s commutes with inductive limits.

4) If u^s is left exact, the natural diagram comparing γ -objects with the underlying sheaves of sets,

$$\begin{array}{ccc} \widetilde{C}_\gamma & \xrightarrow{u_\gamma^s} & \widetilde{C}'_\gamma \\ \downarrow & & \downarrow \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}' \end{array}$$

is commutative up to isomorphism, and u_γ^s is exact.

5) Suppose that the forgetful functor on γ -objects (respectively on C') has a left adjoint $\text{Lib}^\sim$ (respectively $\text{Lib}'^\sim$); cf. Exposé II, 6.5. Then there is a canonical isomorphism

$$u_\gamma^s \text{Lib}^\sim \xrightarrow{\sim} \text{Lib}'^\sim u^s.$$

Moreover, when C is not necessarily small but is a $\mathcal{U}$ -site, the functor u_s^γ still has a left adjoint u_γ^s . Finally, if $\mathcal{V}$ is a universe containing $\mathcal{U}$, and if $u_{\gamma\mathcal{U}}^s$ (respectively $u_{\gamma\mathcal{V}}^s$) denotes the left adjoint to u_s relative to the universe $\mathcal{U}$ (respectively $\mathcal{V}$), the diagram analogous to (1.5.1)

$$\begin{array}{ccc} \widetilde{C}_{\gamma\mathcal{U}} & \xrightarrow{u_{\gamma\mathcal{U}}^s} & \widetilde{C}'_{\gamma\mathcal{U}} \\ \downarrow & & \downarrow \\ \widetilde{C}_{\gamma\mathcal{V}} & \xrightarrow{u_{\gamma\mathcal{V}}^s} & \widetilde{C}'_{\gamma\mathcal{V}} \end{array} \tag{1.7.1}$$

is commutative up to canonical isomorphism.

Proof. The proof in the case where C is a small site is left to the reader. It will be completed in 4.3 in the case where C is a $\mathcal{U}$ -site. $\square$

Notation 1.8. From now on we shall use the notation u_s also for the functor u_s^γ . This notation creates no risk of confusion, since the functor u_s , defined on sheaves of sets, commutes with finite projective limits.

Similarly, when u^s is left exact, we shall use the notation u^s for the functor u_γ^s . This abuse of notation is justified by Proposition 1.7(4).

Example 1.9.1. Let X be a small topological space. Denote by $\text{Open}(X)$ the category of open subsets of X , whose morphisms are inclusions, endowed with the canonical topology. A family $(U_i \subset U)_{i \in I}$ is covering if and only if the open sets U_i cover U (Exposé II, 2.6). The sheaves of sets on $\text{Open}(X)$ are therefore the sheaves of sets in the sense of Godement's *Théorie des faisceaux*. Let $f : X \rightarrow Y$ be a continuous map. It gives a functor

$$f^{-1} : \text{Open}(Y) \longrightarrow \text{Open}(X),$$

which sends an open $U \subset Y$ to its inverse image under f . The functor f^{-1} is continuous (Proposition 1.6). The functor

$$f_s^{-1} : \text{Open}(X)^\sim \longrightarrow \text{Open}(Y)^\sim$$

is the direct-image functor for sheaves in the sense of Godement; the functor

$$(f^{-1})^s : \text{Open}(Y)^\sim \longrightarrow \text{Open}(X)^\sim$$

is the inverse-image functor for sheaves in that same sense. The functor $(f^{-1})^s$ commutes with finite projective limits by Proposition 1.3(5).

Example 1.9.2. Let X be a topological space and let $Y \hookrightarrow X$ be an open subspace. Every open subset of Y is an open subset of X , giving a functor

$$\Phi : \text{Open}(Y) \longrightarrow \text{Open}(X).$$

The functor Φ is continuous (Proposition 1.6). The functor $\Phi_s : \text{Open}(X)^\sim \rightarrow \text{Open}(Y)^\sim$ is the restriction-to- Y functor. The functor

$$\Phi_{\mathbf{Ab}}^s : \text{Open}(Y)_{\mathbf{Ab}}^\sim \longrightarrow \text{Open}(X)_{\mathbf{Ab}}^\sim$$

(Exposé II, 6.3.3 and Exposé I, 1.7) is the extension-by-zero-outside- Y functor of *Théorie des faisceaux*. The functor $\Phi^s : \text{Open}(Y)^\sim \rightarrow \text{Open}(X)^\sim$ is analogous to extension by zero: for every sheaf H on Y and every point $y \in Y$, the stalk at y of $\Phi^s H$ is canonically isomorphic to the stalk at y of H ; for every point $x \in X$, $x \notin Y$, the stalk at x of $\Phi^s H$ is empty. The functor Φ^s is called *extension by the empty set outside Y* . It commutes with fiber products and with finite products over nonempty index sets, but it does not carry the final object of $\text{Open}(Y)^\sim$ to the final object of $\text{Open}(X)^\sim$.

Example 1.9.3. Let C and C' be two small sites, and let $u : C \rightarrow C'$ be a functor between the underlying categories which sends every covering family of C to a covering family of C' . In general u is not continuous (cf. nevertheless Proposition 1.6). Here is a counterexample. Let S be an infinite set of the universe $\mathcal{U}$, and let C be the full subcategory of the category of sets whose objects are the finite nonempty subsets of S . Endow C with the canonical topology. Covering families in C are precisely the jointly surjective families of maps, and no covering family is empty. Moreover, up to isomorphism there are only two constant sheaves on C : the sheaf with value the empty set, and the sheaf with value a singleton. Put $C = C'$, and let $u : C \rightarrow C'$ be a constant functor. The functor u sends covering families of C to covering families of C' , but u is not continuous. Indeed, since the representable presheaves on C' are sheaves, for every object X of C' there is a sheaf F on C' such that $\text{card } F(X) \geq 2$; hence the presheaf $u^* F$ is not a sheaf.

2 Cocontinuous functors

Definition 2.1. Let C and C' be two $\mathcal{U}$ -sites, and let $u : C \rightarrow C'$ be a functor between the underlying categories. We say that u is *cocontinuous* if it has the following property:

(COC) For every object Y of C , and every covering sieve $R \hookrightarrow u(Y)$, the sieve of Y generated by the arrows $Z \rightarrow Y$ such that $u(Z) \rightarrow u(Y)$ factors through R , covers Y .

Proposition 2.2. Let C be a small site, let C' be a $\mathcal{U}$ -site, and let $u : C \rightarrow C'$ be a functor between the underlying categories. Let

$$\hat{u}_* : \hat{C} \longrightarrow \hat{C}'$$

be the functor right adjoint to $F \mapsto F \circ u = \hat{u}^* F$ (Exposé I, 5.1). The functor u is cocontinuous if and only if, for every sheaf G on C , $\hat{u}_* G$ is a sheaf on C' .

Proof. Sufficiency. Suppose that for every sheaf G on C , the presheaf $\hat{u}_* G$ is a sheaf. Let Y be an object of C , and let $R \hookrightarrow u(Y)$ be a covering sieve. One has an isomorphism

$$\mathrm{Hom}(u(Y), \hat{u}_* G) \xrightarrow{\sim} \mathrm{Hom}(R, \hat{u}_* G),$$

and hence, by adjunction,

$$\mathrm{Hom}(\hat{u}^* u(Y), G) \xrightarrow{\sim} \mathrm{Hom}(\hat{u}^* R, G).$$

Thus the morphism $\hat{u}^* R \rightarrow \hat{u}^* u(Y)$ is biconverging (Exposé II, 5.3). Since $\hat{u}^*$ commutes with projective limits, this morphism is also a monomorphism, hence a covering monomorphism. On the other hand, there is a canonical morphism

$$Y \xrightarrow{p} \hat{u}^* u(Y)$$

deduced from the adjunction morphism $\mathrm{id}_C \rightarrow \hat{u}^* u$ (Exposé I, 5.4(3)). Base-changing by p , we obtain a covering sieve of Y :

$$\hat{u}^* R \times_{\hat{u}^* u(Y)} Y \longrightarrow Y.$$

The arrows $Z \rightarrow Y$ which factor through this sieve are exactly those described in (COC).

Necessity. Let S be an object of C' , and let $R \hookrightarrow S$ be a covering sieve. We must prove that, for every sheaf G on C , one has an isomorphism

$$\mathrm{Hom}(S, \hat{u}_* G) \xrightarrow{\sim} \mathrm{Hom}(R, \hat{u}_* G).$$

Using Exposé II, 5.3 and the adjunction isomorphisms, it is enough to prove that the monomorphism $\hat{u}^* R \rightarrow \hat{u}^* S$ is covering. For this it is enough (Exposé II, 5.1) to prove that, for every base change $Y \rightarrow \hat{u}^* S$, with Y an object of C , the sieve

$$\hat{u}^* R \times_{\hat{u}^* S} Y \longrightarrow Y$$

is covering. By the properties of adjoint functors and by Exposé I, 5.4(3), the morphism $Y \rightarrow \hat{u}^* S$ factors through

$$Y \xrightarrow{p} \hat{u}^* u(Y) \longrightarrow \hat{u}^* S.$$

Thus the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y \hookrightarrow Y$ is the base change, along p , of the monomorphism

$$\hat{u}^* R \times_{\hat{u}^* S} \hat{u}^* u(Y) \longrightarrow \hat{u}^* u(Y).$$

Since $\hat{u}^*$ commutes with projective limits, this monomorphism is obtained by applying $\hat{u}^*$ to the covering sieve $R \times_S u(Y) \hookrightarrow u(Y)$. The hypothesis (COC) then says precisely that the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y \hookrightarrow Y$ is covering. $\square$

Proposition 2.3. Let C and C' be two $\mathcal{U}$ -sites, and let $u : C \rightarrow C'$ be cocontinuous. Define

$$u^* : \widetilde{C}' \longrightarrow \widetilde{C}, \quad u^* = \underline{a} \hat{u}^* i',$$

where $\underline{a}$ denotes sheafification for $\hat{C}$, $\hat{u}^*$ is the functor $F \mapsto F \circ u$, and i' is the inclusion $\widetilde{C}' \rightarrow \hat{C}'$.

- 1) The functor u^* commutes with inductive limits and is exact.
- 2) There is a canonical isomorphism $u^*\underline{a}' \simeq \underline{a}\widehat{u}^*$, where $\underline{a}'$ denotes sheafification for $\widehat{C}'$.
- 3) The functor u^* has a right adjoint

$$u_* : \widetilde{C} \longrightarrow \widetilde{C}'.$$

When C is small, the diagram

$$\begin{array}{ccc} \widetilde{C} & \xrightarrow{u_*} & \widetilde{C}' \\ \downarrow i & & \downarrow i' \\ \widehat{C} & \xrightarrow{\widehat{u}_*} & \widehat{C}' \end{array} \quad (2.3.1)$$

where the bottom functor is right adjoint to $\widehat{u}^*$, is commutative up to canonical isomorphism.

- 4) Let $\mathcal{V} \supset \mathcal{U}$ be a universe. If $u_{*\mathcal{U}} : \widetilde{C}_{\mathcal{U}} \rightarrow \widetilde{C}'_{\mathcal{U}}$ and $u_{*\mathcal{V}} : \widetilde{C}_{\mathcal{V}} \rightarrow \widetilde{C}'_{\mathcal{V}}$ denote the right adjoints relative to $\mathcal{U}$ and $\mathcal{V}$, then the diagram

$$\begin{array}{ccc} \widetilde{C}_{\mathcal{U}} & \xrightarrow{u_{*\mathcal{U}}} & \widetilde{C}'_{\mathcal{U}} \\ \downarrow & & \downarrow \\ \widetilde{C}_{\mathcal{V}} & \xrightarrow{u_{*\mathcal{V}}} & \widetilde{C}'_{\mathcal{V}} \end{array} \quad (2.3.2)$$

is commutative up to isomorphism.

Proof. Assertions (1) and (2) follow immediately from Exposé II, 3.4. Assertion (3), when C is small, follows from Proposition 2.2; when C is a $\mathcal{U}$ -site, it will be proved in 4.4. Assertion (4), when C is small, follows from the analogous commutativity assertion when the topologies on C and C' are chaotic, together with Exposé I, 3.5. For chaotic topologies the commutativity of (2.3.2) is immediate from the explicit description of u_* in Exposé I, 5.1. $\square$

2.4

We shall not develop here the considerations concerning γ -objects in categories of sheaves. It will be enough to observe that the functors u^* and u_* introduced in this section always commute with finite projective limits (unlike the functor u^s in the preceding section). Consequently they extend naturally to functors defined on γ -objects; these functors are adjoint to one another and commute with the underlying-sheaf-of-sets functors. We shall use the abuses of notation indicated in 1.8, writing u^* and u_* also for the extensions to γ -objects.

Proposition 2.5. Let C and C' be two $\mathcal{U}$ -sites, and let

$$C \begin{array}{c} \xrightarrow{v} \\ \xleftarrow{u} \end{array} C'$$

be a pair of functors, where v is left adjoint to u . The following properties are equivalent:

- i) The functor u is continuous.
- ii) The functor v is cocontinuous.

Moreover, under these equivalent conditions, there are canonical isomorphisms

$$v_* \simeq u_s, \quad v^* \simeq u^s.$$

In particular, u^s commutes with finite projective limits.

Proof. The property of a functor being continuous or cocontinuous does not depend on the universe $\mathcal{U}$ for which C and C' are $\mathcal{U}$ -sites. This is immediate for a cocontinuous functor, and follows from Proposition 1.5 for a continuous functor. Enlarging the universe if necessary, we may therefore assume C and C' small. The proposition then follows from Proposition 2.2 and Exposé I, 5.6. $\square$

Proposition 2.6. Let $u : C \rightarrow C'$ be a functor which is both continuous and cocontinuous between two $\mathcal{U}$ -sites. Then the functor

$$u^* : \widetilde{C'} \longrightarrow \widetilde{C}$$

(Proposition 2.3) commutes with $\mathcal{U}$ -inductive and projective limits. Let

$$u_! : \widetilde{C} \longrightarrow \widetilde{C'}, \quad u_* : \widetilde{C} \longrightarrow \widetilde{C'}$$

be the left and right adjoints to u^* , respectively. The functor $u_!$ is fully faithful if and only if u_* is fully faithful. When u is fully faithful, $u_!$ is fully faithful; conversely, if the topologies of C and C' are coarser than the canonical topologies, full faithfulness of $u_!$ implies full faithfulness of u .

Proof. The functor u^* has a right adjoint u_* (Proposition 2.3) and a left adjoint $u_!$ (Proposition 1.2). Hence u^* commutes with inductive and projective limits (Exposé I, 2.11). The assertion that $u_!$ is fully faithful if and only if u_* is fully faithful is a general property of adjoint functors (Exposé I, 5.7(1)). If u is fully faithful, then for a sufficiently large universe $\mathcal{V}$, the functor

$$\widehat{u}_* : \widehat{C}_{\mathcal{V}} \longrightarrow \widehat{C'}_{\mathcal{V}}$$

on $\mathcal{V}$ -presheaves is fully faithful (Exposé I, 5.7). Hence $u_* : \widetilde{C} \rightarrow \widetilde{C'}$ is fully faithful by the commutativity of (2.3.1) and (2.3.2), and therefore $u_!$ is fully faithful by the preceding paragraph. The converse follows from the commutative diagram of Proposition 1.2(iv), together with the fact that the functors ϵ_C and $\epsilon_{C'}$ are fully faithful when the topologies are coarser than the canonical topologies. $\square$

Example 2.7. With the notation of Example 1.9.2, the functor

$$\Phi : \text{Open}(Y) \longrightarrow \text{Open}(X)$$

is continuous and cocontinuous. Moreover, let $i : Y \rightarrow X$ be the canonical inclusion and let

$$i^{-1} : \text{Open}(X) \longrightarrow \text{Open}(Y)$$

be the functor it defines (Example 1.9.1). The functor i^{-1} is right adjoint to Φ . Thus one has a chain of three adjoint functors

$$\Phi^s, \quad \Phi_s = \Phi^*, \quad \Phi_*,$$

and canonical isomorphisms

$$\Phi^* \simeq (i^{-1})^s, \quad \Phi_* \simeq i_s^{-1}.$$

3 Induced topology

3.1

Let C' be a site, let C be a category, and let $u : C \rightarrow C'$ be a functor. For every universe $\mathcal{U}$ such that C' is a $\mathcal{U}$ -site and C is a $\mathcal{U}$ -small category, let $\mathcal{T}_{\mathcal{U}}$ be the finest among the topologies T on C making u continuous (Definition 1.1). Such a topology exists by Exposé II, 2.2. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$: if $\mathcal{V} \supset \mathcal{U}$, then $\mathcal{T}_{\mathcal{U}} = \mathcal{T}_{\mathcal{V}}$, by Definition 1.1 and Proposition 1.5. This topology is called the *topology induced on C by the topology of C' via the functor u* .⁴

Proposition 3.2. Let C be a small category, let C' be a $\mathcal{U}$ -site, let $u : C \rightarrow C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by u . Let X be an object of C , and let $R \hookrightarrow X$ be a sieve on X . The following properties are equivalent:

- i) The sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$.
- ii) For every base change $Y \rightarrow X$, where Y is an object of C , the morphism

$$u_!(R \times_X Y) \longrightarrow u(Y)$$

is biconverging in $\widehat{C'}$ (Exposé II, 5.2).

Proof. The implication i) $\Rightarrow$ ii) follows from axiom (T1) for topologies and from Proposition 1.2. Conversely, for every sheaf F on C' , the map

$$\mathrm{Hom}(Y, u^*F) \longrightarrow \mathrm{Hom}(R \times_X Y, u^*F)$$

is isomorphic by adjunction to

$$\mathrm{Hom}(u(Y), F) \longrightarrow \mathrm{Hom}(u_!(R \times_X Y), F),$$

which is bijective by Exposé II, 5.3. Hence, by Exposé II, 2.2, the sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$. $\square$

Corollary 3.3. Let C' be a site, let C be a category, let $u : C \rightarrow C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by the topology of C' . Let $(X_i \rightarrow X)_{i \in I}$ be a family of base-changeable morphisms in C , and suppose that u commutes with the relevant fiber products.⁵ The following properties are equivalent:

- i) The family $(X_i \rightarrow X)_{i \in I}$ is covering for $\mathcal{T}$.
- ii) The family $(u(X_i) \rightarrow u(X))_{i \in I}$ is covering in C' .

Proof. The implication i) $\Rightarrow$ ii) follows from Proposition 1.6. Conversely, choose a universe $\mathcal{U}$ such that C is $\mathcal{U}$ -small and C' is a $\mathcal{U}$ -site. Let $R \hookrightarrow X$ be the sieve generated by the $X_i \rightarrow X$. The presheaf R is the coequalizer of

$$\coprod_{i,j} X_i \times_X X_j \rightrightarrows \coprod_i X_i$$

⁴When there is no risk of confusion, it is simply called the topology induced on C by the topology of C' .

⁵In fact it is enough that u commute with fiber products of the form $X_i \times_X X'$.

(the coproduct being taken in $\widehat{C}$); cf. Exposé I, 2.12 and 3.5. Since $u_!$ commutes with inductive limits, $u_!R$ is the coequalizer of

$$\coprod_{i,j} u(X_i \times_X X_j) \rightrightarrows \coprod_i u(X_i).$$

As u commutes with fiber products, $u_!R$ is the coequalizer of

$$\coprod_{i,j} u(X_i) \times_{u(X)} u(X_j) \rightrightarrows \coprod_i u(X_i).$$

Thus $u_!R \rightarrow u(X)$ is the sieve on $u(X)$ generated by the morphisms $u(X_i) \rightarrow u(X)$. Using again that u commutes with fiber products, one proves that for every base change $Y \rightarrow X$, with Y an object of C , the morphism

$$u_!(R \times_X Y) \longrightarrow u(Y)$$

is the sieve on $u(Y)$ generated by the morphisms

$$u(X_i) \times_{u(X)} u(Y) \longrightarrow u(Y).$$

Since the family $u(X_i) \rightarrow u(X)$ is covering, the family $u(X_i) \times_{u(X)} u(Y) \rightarrow u(Y)$ is covering. Hence $u_!(R \times_X Y) \rightarrow u(Y)$ is a covering sieve, and by Proposition 3.2 the sieve $R \rightarrow X$ is covering for $\mathcal{T}$. $\square$

Corollary 3.4. Let C' be a $\mathcal{U}$ -site, let C be a full subcategory of C' , and let $u : C \rightarrow C'$ be the inclusion functor. Suppose fiber products are representable in C , and that u commutes with fiber products. The following conditions are equivalent:

- i) a) For every object X of C , every covering family $(Y_j \rightarrow X)_{j \in J}$ in C' is refined by a covering family $(X_i \rightarrow X)_{i \in I}$ with X_i objects of C .
- b) There exists a small set G of objects of C such that every object of C is the target of a covering family in C' whose sources are in G .
- ii) The topology induced by the topology of C' is a $\mathcal{U}$ -topology, and u is both continuous and cocontinuous for this topology.

Proof. This follows immediately from Corollary 3.3 and Definition 2.1. $\square$

Proposition 3.5. Let C be a $\mathcal{U}$ -site and let $\epsilon_C : C \rightarrow \widetilde{C}$ be the canonical functor (Exposé II, 4.4.0). Endow $\widetilde{C}$ with the canonical topology. Then the topology of the site C is the topology induced by the topology of $\widetilde{C}$.

Proof. The covering families for the canonical topology on $\widetilde{C}$ are the universally effective epimorphic families (Exposé II, 2.5), i.e., by Exposé II, 4.3, the epimorphic families. Let T be the topology of the site C , and let $\mathcal{T}_{\mathcal{U}}$ be the finest topology T' on C such that, for every sheaf F on $\widetilde{C}$, the composite $F \circ \epsilon_C$ is a sheaf for T' . It is enough to show that $T = \mathcal{T}_{\mathcal{U}}$, for then $\mathcal{T}_{\mathcal{U}}$ is a $\mathcal{U}$ -topology and hence is the induced topology.

A) *The topology T is finer than $\mathcal{T}_{\mathcal{U}}$.* Let $(X_i \rightarrow X)_{i \in I}$ be a covering family for $\mathcal{T}_{\mathcal{U}}$. Then for every sheaf F on $\widetilde{C}$, the map

$$F(\epsilon_C X) \longrightarrow \prod_i F(\epsilon_C X_i)$$

is injective. Thus $(\epsilon_C X_i \rightarrow \epsilon_C X)_{i \in I}$ is covering for the canonical topology of $\tilde{C}$, and therefore $(X_i \rightarrow X)_{i \in I}$ is covering for T .

B) *The topology T is coarser than $\mathcal{T}_{\mathcal{U}}$.* It is enough to prove that $\epsilon_C : C \rightarrow \tilde{C}$ is continuous. In Exposé IV, Section 1, we shall prove that every sheaf on $\tilde{C}$ is representable; hence, for every sheaf F on $\tilde{C}$, the composite $F \circ \epsilon_C$ is a sheaf on C . Proposition 3.5 will not be used before Exposé IV, Section 1.

We record two results that will be used in Exposé VI, Section 7. □

Proposition 3.6. Let $(C_i)_{i \in I}$ be a family of sites, let C be a category, and for each $i \in I$ let $u_i : C_i \rightarrow C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$ -small. There exists a topology $\mathcal{T}_{\mathcal{U}}$ on C which is the coarsest topology for which all the u_i are continuous. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories under consideration are small.

The last assertion of Proposition 3.6 follows from Proposition 1.5. Let T be a topology on C . Then the functors u_i are continuous if and only if, for every $i \in I$ and every covering sieve $R \hookrightarrow X$ of an object X of C_i , the morphism

$$u_{i!}(R) \longrightarrow u_i(X)$$

in $\hat{C}$ is bicobering (Proposition 1.2(ii)). Thus Proposition 3.6 follows from the following lemma.

Lemma 3.6.1. Let C be a small category, and let $(u_i : F_i \rightarrow G_i)_{i \in I}$ be a family of arrows in $\hat{C}$. Then there exists on C a coarsest topology among those which make the morphisms u_i covering (respectively bicobering).

To say that $u : F \rightarrow G$ is covering for a given topology T means that for every arrow $X \rightarrow G$, with $X \in \text{Ob } C$, the corresponding arrow $F \times_G X \rightarrow X$ is covering; equivalently, the family of arrows $X' \rightarrow X$ in C which factor through this arrow is covering. The existence of a coarsest topology for which the $u_i : F_i \rightarrow G_i$ are covering follows from Exposé I, 1.1.6, proving the non-parenthetical assertion of Lemma 3.6.1. The bicobering assertion follows by recalling that a morphism $u : F \rightarrow G$ is bicobering if and only if the two morphisms $u : F \rightarrow G$ and $\text{diag}_u : F \rightarrow F \times_G F$ are covering.

Proposition 3.7. Let $(C_i)_{i \in I}$ be a family of sites, let C be a category, for each $i \in I$ let $u_i : C_i \rightarrow C$ be a functor, and let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$ -small. There exists a topology $\mathfrak{T}_{\mathcal{U}}$ which is the finest topology for which all the u_i are cocontinuous. The topology $\mathfrak{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories under consideration are small.

For a universe $\mathcal{U}$ in which the categories under consideration are small, the functors u_i are cocontinuous for a topology $\mathfrak{T}$ on C if and only if, for every $i \in I$ and every sheaf F on C_i , the presheaf $\hat{u}_{i*}F$ is a sheaf for $\mathfrak{T}$ (Proposition 2.2). It follows that $\mathfrak{T}_{\mathcal{U}}$ is the finest topology for which the presheaves $\hat{u}_{i*}F$, $i \in I$, $F \in \text{Ob } C_i$, are sheaves (Exposé II, 2.2). The last assertion follows from Proposition 2.2.

4 The comparison lemma

Theorem 4.1 (Comparison lemma). Let C be a small category, let C' be a site whose underlying category is a $\mathcal{U}$ -category, and let $u : C \rightarrow C'$ be a fully faithful functor. Endow C with the topology induced by u (3.1). Consider the following properties:

- i) Every object of C' can be covered by objects coming from C .
- ii) The functor $F \mapsto F \circ u$ induces an equivalence from the category of sheaves on C' to the category of sheaves on C .

One always has i) $\Rightarrow$ ii). If C' is a $\mathcal{U}$ -site and the topology of C' is coarser than the canonical topology, then ii) $\Rightarrow$ i).

4.1.1

We first prove i) $\Rightarrow$ ii). The proof has two steps.

First step. For every presheaf H on C' , the adjunction morphism

$$\varphi : u_! u^* H \longrightarrow H$$

(Exposé I, 5.1) is bicovering (Exposé II, 5.3), and the functor $u_s : \widetilde{C}' \rightarrow \widetilde{C}$ is fully faithful.

Let C/H be the small category whose objects are pairs consisting of an object Y of C and a morphism $u(Y) \rightarrow H$, and whose morphisms are the commutative triangles

$$\begin{array}{ccc} u(Y) & \xrightarrow{u(m)} & u(Y') \\ & \searrow & \downarrow \\ & & H. \end{array}$$

One has

$$u^* H = \varinjlim_{Y \in \text{Ob}(C/H)} Y$$

(Exposé I, 3.4), and hence

$$u_! u^* H = \varinjlim_{Y \in \text{Ob}(C/H)} u(Y)$$

(Exposé I, 5.4). The adjunction morphism is the evident morphism resulting from this description of $u_! u^* H$ as an inductive limit. It has the following property: (*) for every object Y of C , every morphism $m : u(Y) \rightarrow H$ factors uniquely as the canonical morphism $\alpha(m) : u(Y) \rightarrow u_! u^* H$ followed by φ .

It follows immediately from i) that φ is covering. We prove it is bicovering. Let $p, q : Z \rightrightarrows u_! u^* H$ be two morphisms from an object Z of C' such that $\varphi p = \varphi q$. For every object Y of C and every morphism $n : u(Y) \rightarrow Z$, one has $\varphi p n = \varphi q n$. Property (*) then implies $p n = q n$. Since the objects $u(Y)$ cover Z , the equalizer of (p, q) is a covering sieve of Z . Hence φ is bicovering.

For every sheaf H on C' , $u^* H$ is a sheaf on C , denoted $u_s H$. Moreover

$$u^s u_s H = \underline{a}' u_! u_s H$$

by Proposition 1.3, and the adjunction morphism $u^s u_s H \rightarrow H$ is obtained by applying sheafification to $\varphi : u_! u^* H \rightarrow H$. Therefore $u^s u_s H \rightarrow H$ is an isomorphism. Thus $u_s : \widetilde{C}' \rightarrow \widetilde{C}$ is fully faithful.

Second step. The functor u is cocontinuous, and the functor $u^s : \widetilde{C} \rightarrow \widetilde{C}'$ of Proposition 1.2 is fully faithful. Hence $u_s : \widetilde{C}' \rightarrow \widetilde{C}$ is an equivalence.

Let Y be an object of C , and let $i : R \hookrightarrow u(Y)$ be a covering sieve. Since u^* commutes with projective limits (Exposé I, 5.5), the morphism $u^*(i) : u^*(R) \hookrightarrow u^*u(Y)$ is a monomorphism. Since u is fully faithful, $u^*u(Y) \simeq Y$, hence $u^*(i)$ is a sieve of Y . To prove that u is cocontinuous, it is enough to show that $u^*(i) : u^*(R) \hookrightarrow Y$ is covering for the induced topology on C . By Proposition 3.2, it is enough to show that for every base change $m : X \rightarrow Y$, the morphism

$$u_!(u^*(R) \times_Y X) \longrightarrow u(X)$$

is bicoverting. But u^* commutes with projective limits, so

$$u^*(R) \times_Y X = u^*(R \times_{u(Y)} u(X)),$$

and $R \times_{u(Y)} u(X)$ is a covering sieve of $u(X)$. It is therefore enough to show that, for every object Y of C and every covering sieve $i : R \hookrightarrow u(Y)$, the morphism in $\widehat{C'}$

$$u_!u^*(R) \xrightarrow{u_!u^*(i)} u(Y)$$

is bicoverting. This morphism factors as the adjunction morphism $u_!u^*(R) \rightarrow R$, which is bicoverting by the first step, followed by the covering monomorphism $R \hookrightarrow u(Y)$. It is therefore bicoverting. This proves that u is cocontinuous. Since u is continuous and fully faithful, Proposition 2.6, used in the case where C is small, implies that u^s (denoted $u_!$ in Proposition 2.6) is fully faithful. Since u^s and u_s are adjoint and both fully faithful, they are quasi-inverse functors; hence u_s is an equivalence.

4.1.2

We now prove ii) $\Rightarrow$ i). The objects Y of C form a generating family of $\widetilde{C}$ (Exposé II, 4.10). Hence, for every object X of C' , there exists an epimorphic family in $\widetilde{C}$

$$v_i : Y_i \longrightarrow u_s X, \quad i \in I.$$

Since $u_s u(Y_i) = Y_i$, and since u_s is an equivalence of categories, one has $v_i = u_s(w_i)$, where the morphisms $w_i : u(Y_i) \rightarrow X$ form an epimorphic family in $\widetilde{C'}$. It follows from Exposé II, 5.1 that the family $w_i : u(Y_i) \rightarrow X$, $i \in I$, is covering.

4.1 Completion of the proof of Proposition 1.2

Let G be a full subcategory of C whose objects form a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \rightarrow C$. The functor

$$(u \circ i)_s = i_s \circ u_s$$

has a left adjoint by the first part of the proof of Proposition 1.2. Since i_s is an equivalence of categories by Theorem 4.1, the functor u_s has a left adjoint u^s , and there is a functorial isomorphism

$$u^s \simeq (u \circ i)^s \circ i_s.$$

We prove that the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ \downarrow \epsilon_C & & \downarrow \epsilon_{C'} \\ \widetilde{C} & \xrightarrow{u^s} & \widetilde{C'} \end{array} \quad (4.2.1)$$

is commutative up to isomorphism. For every sheaf H on C' , one has

$$\mathrm{Hom}_{\widetilde{C}'}(\epsilon_{C'}uX, H) \simeq u_s H(X)$$

for every $X \in \mathrm{Ob} C$. Moreover,

$$\mathrm{Hom}_{\widetilde{C}'}(u^s \epsilon_C X, H) \simeq \mathrm{Hom}_{\widetilde{C}}(\epsilon_C X, u_s H) \simeq u_s H(X),$$

first by adjunction and then by the definition of ϵ_C . Thus $\epsilon_{C'}uX \simeq u^s \epsilon_C X$ for every $X \in \mathrm{Ob} C$. This proves i) $\Rightarrow$ iv). Conversely, assume iv). One has a commutative diagram

$$\begin{array}{ccccccc} G & \xrightarrow{i} & C & \xrightarrow{u} & C' & & \\ \downarrow \epsilon_G & & \downarrow \epsilon_C & & \downarrow \epsilon_{C'} & & \\ \widetilde{G} & \xrightarrow{i^s} & \widetilde{C} & \xrightarrow{u^s} & \widetilde{C}' & & \end{array} \quad (4.2.2)$$

It follows, by the first part of the proof, that $u \circ i : G \rightarrow C'$ is continuous. Theorem 4.1 and Definition 1.1 then immediately imply that u is continuous. This proves iv) $\Rightarrow$ i). The uniqueness of u^s follows as well, since it is already known in the small case.

4.2 Completion of the proof of Propositions 1.5 and 1.7

Let $u : C \rightarrow C'$ be a functor between two $\mathcal{U}$ -sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \rightarrow C$. It follows immediately from Theorem 4.1 and Definition 1.1 that u is continuous if and only if $u \circ i$ is continuous. From this remark and from the first part of the proof of Proposition 1.5, the general case follows. The same remark also reduces the proof of Proposition 1.7 to the case where C is small.

4.3 Completion of the proof of Proposition 2.3

Let $u : C \rightarrow C'$ be a functor between two $\mathcal{U}$ -sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1), endowed with the topology induced by the inclusion functor $i : G \rightarrow C$. It follows from the proof of Theorem 4.1, specifically the second step of 4.1.1, that i is cocontinuous. Hence, if u is cocontinuous, the functor $u \circ i : G \rightarrow C'$ is cocontinuous. Moreover

$$i^* : \widetilde{C} \longrightarrow \widetilde{G}$$

is an equivalence of categories by Theorem 4.1. Therefore the functor $u^* : \widetilde{C}' \rightarrow \widetilde{C}$ has a right adjoint. This proves Proposition 2.3(3). To prove (4), it is enough to observe that

$$(u \circ i)_* \simeq u_* \circ i_*,$$

and that i_* is an equivalence (Theorem 4.1). The commutativity of (2.3.2) then follows from the commutativity of these diagrams when C is small.

5 Localization

5.1

Let now C be a $\mathcal{U}$ -site and let X be an object of $\widehat{C}$. Unless explicitly stated otherwise, the category C/X will be endowed with the topology $\mathcal{T}$ induced by the functor

$$j_X : C/X \longrightarrow C.$$

The notation C/X will denote the category C/X endowed with this topology. Proposition 5.11 of Exposé I shows that the functor $j_{X!}$ commutes with fiber products, hence sends monomorphisms to monomorphisms. In particular, for every object $(Y \rightarrow X)$ of C/X , the functor $j_{X!}$ establishes a one-to-one correspondence between sieves, in C/X , of $(Y \rightarrow X)$, and sieves, in C , of Y .

Proposition 5.2. Let C be a $\mathcal{U}$ -site, let X be an object of $\widehat{C}$, and let $j_X : C/X \rightarrow C$ be the corresponding continuous functor.

- 1) A sieve R of an object $(Z \rightarrow X)$ is covering in C/X if and only if the sieve $j_{X!}(R) \hookrightarrow Z$ is covering in C .
- 2) The functor $j_X : C/X \rightarrow C$ is both cocontinuous and continuous.
- 3) Let $m : Y \rightarrow X$ be a morphism in $\widehat{C}$. Then one has a commutative diagram

$$\begin{array}{ccc} C/Y & \xrightarrow{j_m} & C/X \\ & \searrow j_Y & \downarrow j_X \\ & & C. \end{array}$$

The topology induced on C/Y by j_Y is equal to the topology induced on C/Y by j_m .

- 4) The topology induced by $j_X : C/X \rightarrow C$ is a $\mathcal{U}$ -topology (Exposé II, 3.0.2).

Proof.

- 1) If the sieve R of $(Z \rightarrow X)$ is covering, then $j_{X!}(R) \hookrightarrow Z$ is covering by Proposition 1.6. Conversely, if the sieve $j_{X!}(R) \rightarrow Z$ is covering in C , then the same is true after every base change in C/X . Hence R is covering by Proposition 3.2.
- 2) This follows immediately from (1), applying Definition 2.1.
- 3) This follows immediately from the description of covering sieves given in (1).
- 4) Let $(G_i)_{i \in I}$ be a small topologically generating family of C . One checks immediately that the small family

$$(u : G_i \rightarrow X), \quad u \in \coprod_{i \in I} \text{Hom}_{\widehat{C}}(G_i, X),$$

is topologically generating in C/X . □

Terminology and notation 5.3. By the preceding proposition, the functor j_X is both continuous and cocontinuous. It therefore defines a sequence of three adjoint functors between categories of sheaves of sets (Propositions 1.3 and 2.3):

$$j_X^s : (C/X)^\sim \longrightarrow \widetilde{C}, \quad j_X^* = j_{X,s} : \widetilde{C} \longrightarrow (C/X)^\sim, \quad j_{X*} : (C/X)^\sim \longrightarrow \widetilde{C}.$$

In the particular situation of Proposition 5.2, we shall use the following terminology and notation:

- 1) The functor j_{X*} will be called the *direct-image functor*.
- 2) The functor $j_{X,s} = j_X^*$ will be denoted j_X^* and called the *restriction-to- C/X functor*.
- 3) The functor j_X^s on sheaves of sets will be called the *extension by the empty set to the category C functor* and will be denoted $j_{X!}$ (cf. 2.9.2).

Thus one has a sequence of three adjoint functors between $(C/X)^\sim$ and $\tilde{C}$:

$$j_{X!}, \quad j_X^*, \quad j_{X*}.$$

Proposition 5.4. The functor

$$j_{X!} : (C/X)^\sim \longrightarrow \tilde{C}$$

factors through the category $\tilde{C}/\underline{a}X$, where $\underline{a}$ is sheafification:

$$(C/X)^\sim \xrightarrow{e_X^\sim} \tilde{C}/\underline{a}X \longrightarrow \tilde{C}.$$

The functor

$$e_X^\sim : (C/X)^\sim \longrightarrow \tilde{C}/\underline{a}X$$

is an equivalence of categories. The restriction-to- C/X functor, composed with the equivalence $e_X^\sim$, is isomorphic to base change along $\underline{a}X \rightarrow \epsilon$, where ϵ is the final object of $\tilde{C}$:

$$F \longmapsto (F \times \underline{a}X \xrightarrow{\text{pr}_2} \underline{a}X).$$

Proof. The image under $j_{X!}$ of the final object of $(C/X)^\sim$ is $\underline{a}X$, whence the factorization. To prove that $e_X^\sim$ is an equivalence, we give only a few indications. By Exposé I, 5.11, a presheaf on C/X is defined by a presheaf F on C together with a morphism $F \rightarrow X$. One then proves that the following two properties are equivalent:

- i) The presheaf on C/X defined by $F \rightarrow X$ is a sheaf.
- ii) The following square is cartesian, where i and $\underline{a}$ denote the inclusion into presheaves and sheafification:

$$\begin{array}{ccc} F & \longrightarrow & i\underline{a}F \\ \downarrow & & \downarrow \\ X & \longrightarrow & i\underline{a}X. \end{array}$$

It follows immediately that $e_X^\sim$ is an equivalence. The last assertion is trivial. $\square$

Proposition 5.5.

- 1) Let C be a $\mathcal{U}$ -site and let X be an object of $\hat{C}$. The following diagram of categories and functors is commutative up to canonical isomorphism:

$$\begin{array}{ccccc} C/X & \rightarrow & \widehat{C/X} & \xrightarrow{\underline{a}_X} & (C/X)^\sim \rightarrow \widehat{C/X} \\ \downarrow & & \downarrow & & \downarrow \downarrow \\ C/X & \rightarrow & \hat{C}/X & \xrightarrow{\underline{a}/X} & \tilde{C}/\underline{a}X \rightarrow \hat{C}/i\underline{a}X \rightarrow \hat{C}/X. \end{array}$$

Here $\underline{a}_X$ and i_X denote sheafification and inclusion into presheaves for the site C/X . The notation $\underline{a}/X$ denotes the natural extension of $\underline{a}$, the sheafification functor for C , to the category of arrows with target X ; similarly for $i/\underline{a}X$. Finally, π denotes the base-change functor along the canonical arrow $X \rightarrow i\underline{a}X$.

- 2) Let, moreover, $m : Y \rightarrow X$ be a morphism of $\widehat{C}$. The following diagram is commutative up to canonical isomorphism:

$$\begin{array}{ccc} \tilde{C}/\underline{a}X/\underline{a}Y & \xleftarrow{f} & (C/X)^\sim/\underline{a}_X Y \\ \parallel & & \uparrow g \\ \tilde{C}/\underline{a}Y & \xleftarrow{e_Y^\sim} & (C/Y)^\sim \xleftarrow{e_m^\sim} (C/X/Y)^\sim. \end{array}$$

The arrow f is the natural extension of the equivalence $e_X^\sim$ to the category of arrows with target $\underline{a}_X Y$, and is therefore an equivalence of categories. The arrow g is precisely the equivalence of Proposition 5.4 applied to the situation $(C/X)/[m]$.

- 3) Let

$$j_{\underline{a}X}! : \tilde{C}/\underline{a}X \longrightarrow \tilde{C}$$

be the functor “forget $\underline{a}X$ ”, and let

$$j_{\underline{a}X}^* : \tilde{C} \longrightarrow \tilde{C}/\underline{a}X$$

be the functor “product with $\underline{a}X$ ”. The diagrams

$$\begin{array}{ccc} (C/X)^\sim & \xrightarrow{j_{X!}} & \tilde{C} \\ \downarrow e_X^\sim & & \parallel \\ \tilde{C}/\underline{a}X & \xrightarrow{j_{\underline{a}X}!} & \tilde{C} \end{array} \quad \text{and} \quad \begin{array}{ccc} \tilde{C} & \xrightarrow{j_X^*} & (C/X)^\sim \\ \parallel & & \downarrow e_X^\sim \\ \tilde{C} & \xrightarrow{j_{\underline{a}X}^*} & \tilde{C}/\underline{a}X \end{array}$$

are commutative up to canonical isomorphism.

Proof. The proof of these assertions is immediate from the definition of the equivalences $e_X^\sim$ of Proposition 5.4 and $\widehat{e}_X$ of Exposé I, 5.11. $\square$

References

- [1] R. Godement, *Théorie des faisceaux*, Hermann, 1958, Act. Scient. Ind. no. 1252, Paris.